DriveCompose: Generative Reconstruction of Compositional Driving Scenes

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Abstract

Reusing a driving scene under new camera and actor states exposes surfaces that were never recorded. Reconstructing such a scene requires recovering the observed content, completing missing geometry and appearance, and retaining a stable identity for each movable component. We present DriveCompose, a framework for generative reconstruction of compositional driving scenes from calibrated image sequences. A shared visual encoder estimates geometry, actor identities, and motion. We aggregate this evidence into world-space background queries and canonical vehicle and human queries, then encode them in a structured latent representation. A conditional flow model completes unknown codes while exchanging context across scene components. Observed codes and known free space constrain this generation process. Type-specific decoders produce Gaussian background and assets, which are checked against the input views before use. Learning proceeds from visual recovery and latent representation fitting to completion-prior training and adaptation with rendered observations. The reconstructed scene is subsequently composed under explicit camera, rigid-body, and skeletal states. This formulation gives generation a persistent three-dimensional output: the same completed content can be rendered along different trajectories without resampling its geometry or appearance.

1. Introduction

Driving datasets record scenes along captured trajectories [3, 49]. Driving simulation and scene editing require a reconstruction that remains usable when the recorded configuration changes. Moving a camera reveals a road patch hidden behind a vehicle. Turning that vehicle exposes its unseen side, and changing a pedestrian's pose brings previously occluded body surfaces into view. These operations require geometry and appearance beyond the recorded images, together with a representation that keeps the content attached to the same scene and actor identities.

Recent feed-forward methods recover driving scenes directly from temporal observations. STORM aggregates dynamic Gaussians through estimated scene flow, and DGGT jointly recovers camera parameters and dynamic scene content from unposed images [5, 71]. StreetForward and PointForward further study temporal processing and the organization of reconstructed evidence [8, 77]. For an asset that will later be moved or articulated, however, observations across time must also be gathered into a stable local representation, and the surfaces exposed by the new state must be supplied.

Compositional representations address the first requirement. Street Gaussians separates vehicles from the background; OmniRe includes canonical models for rigid and articulated actors; HUGSIM connects scene content to control-driven state updates [7, 68, 80]. Generative models address missing content through asset priors, Gaussian completion, or generated views [26, 69, 75, 83]. Combining them requires a generator that assigns content to the background and to individual assets. A generated surface must have an owner, a coordinate frame, and a motion rule before it can become part of an editable scene.

We formulate driving-scene reconstruction as conditional generation of a compositional scene. Our representation follows the units that will be manipulated: a world-space background, rigid vehicles in local coordinates, and humans on canonical surfaces with skeleton bindings. Image evidence is first aggregated on these supports. A generative model then completes their missing latent content under the observed scene conditions. Each completed surface is decoded in the coordinate frame used to control its component.

DriveCompose implements this formulation with a visual recovery front end, structured latent codes, and a conditional flow model (Fig. 1). The front end estimates geometry, trajectory-level identities, and time-varying states. Canonical aggregation combines observations of the same content across views and time. Within the completion model, local attention follows each component's support, while shared memory exchanges context between background and assets. Each query carries both its canonical coordinate and its placement in a reference world configuration: the former identifies persistent content, and the latter expresses its spatial relation to the scene. Type-specific decoders convert the

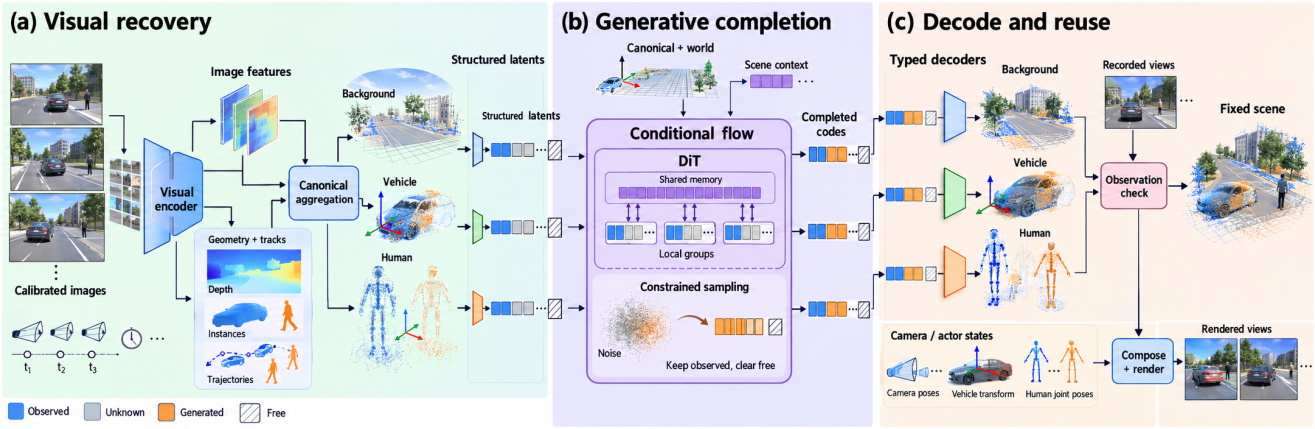


Figure 1. **DriveCompose overview.** (a) Calibrated images provide geometry, identity, and motion evidence, which is aggregated on background and canonical asset supports and encoded into structured latents. (b) A scene-conditioned flow transformer completes unknown codes while retaining observed content and suppressing free space. (c) Typed decoders produce Gaussian components. A check in the original views precedes scene reuse; camera and actor states then control composition and rendering. Scene depictions are schematic, not experimental outputs.

completed codes into Gaussian components.

Generation must also respect the images from which the scene was recovered. We retain reliable observed codes and suppress known free space throughout sampling. Newly generated content can nevertheless occlude an observed surface, so we check the assembled candidate in the original views as well. After construction, the background and canonical assets remain fixed. Camera placement, vehicle transforms, and human joint poses determine their subsequent rendering. Learning follows this separation: we fit recovery and the latent representation before training the completion prior, then adapt generation through differentiable rendering while keeping the representation fixed.

Our contributions are:

- A generative reconstruction framework that organizes driving observations into a compositional Gaussian scene, with world-space background and persistent canonical assets for state-driven reuse.
- A structured completion model that combines category-specific supports and scene-wide context with observed-code constraints, free-space suppression, and rendered observation checks.

2. Related work

Radiance representations. NeRF represents geometry and appearance through a continuous radiance field [33], with extensions for unconstrained photographs and multi-scale sampling [1, 31]. Gaussian splatting provides explicit primitives for rendering [22]; Mip-Splatting addresses aliasing [76], while Gaussian ray tracing and 3DGUT extend ray handling and camera models [34, 63]. D-NeRF and deformable Gaussians introduce time-dependent deformation [40, 74]. These representations establish the rendering machinery on which scene decomposition and asset control can be built.

Urban scenes and control. Urban Radiance Fields and Block-NeRF address large urban captures [43, 50]. NeuRAD, EVolSplat, and SplatAD adapt neural rendering to driving scenes and sensors [17, 32, 53]. Neural Scene Graphs and EmerNeRF organize dynamics through object structure or self-supervised decomposition [35, 70]. Street Gaussians, DrivingGaussian, and holistic HUGS distinguish foreground content from the background [68, 81, 82]; OmniRe extends reconstruction to diverse actor types [7]. UniSim and HUGSIM connect reconstructed scenes to closed-loop simulation [73, 80]. We preserve component identity and motion rules within the variables used for generative completion.

Feed-forward reconstruction. LRM, LGM, and GS-LRM learn image-to-3D prediction [18, 51, 78]. pixelSplat, MVSplat, and DepthSplat recover Gaussians from sparse views [4, 6, 67], while DUSt3R, MASt3R, and VGGT recover geometry and correspondences with transformers [25, 55, 58]. Fast3R and CUT3R explore joint and recurrent processing of image collections [56, 72]. TokenGS explicitly decouples Gaussian prediction from pixels through learnable tokens [44]. Our distinction concerns the variables and inference task: conditional generation completes background and canonical asset codes before they are decoded into a reusable scene.

Driving-scene inference. STORM predicts dynamic Gaussians and aggregates them through scene flow [71]. DrivingForward studies flexible surround-view inputs [52], and DGGT combines pose-free reconstruction, temporal visibility, and motion-guided interpolation [5]. StreetForward, PointForward, and Instant NuRec investigate temporal processing, point-aligned representations, and simulation-oriented reconstruction [8, 21, 77]. DriveCompose uses recovered geometry and motion to gather evidence on persistent supports, where missing content can be completed before state changes expose it.

Canonical and animatable assets. SMPL and SMPL-X define articulated body models [29, 37]; Humans in 4D and WHAM recover human shape and motion from images [15, 48]. HumanNeRF and Vid2Avatar reconstruct people from videos [16, 61], while GaussianAvatar, GauHuman, human HUGS, and 3DGS-Avatar use animatable Gaussians [19, 20, 24, 41]. CanonicalFusion, FRESA, LHM, and IDOL infer drivable or animatable human assets from one or more images [42, 47, 57, 84]. UrbanCAD instead targets controllable vehicle assets [30]. Segmentation models and synthetic body data offer complementary supervision resources [2, 23]. We use category-specific supports to retain each asset's coordinates and motion binding throughout completion.

Generative reconstruction and completion. DreamFusion distills a 2D diffusion prior into 3D content [39]; Zero-1-to-3, MVDream, and CAT3D generate novel or multiple views for 3D creation [12, 28, 46]. Objaverse provides explicit asset data for 3D learning [9]. GenAssets learns driving-asset priors [75], and structured latent models jointly encode geometry and appearance for generation [65, 66]. CompleteSplat completes Gaussian scenes, JRM combines unaligned object observations through latent flow matching, and LatentWorld organizes semantic occupancy generation around entities [26, 36, 64]. DriveCompose completes RGB-renderable background and canonical assets in a shared query layout, with observed codes and free space constraining the unknown content.

Controlled driving-video generation. MagicDrive, MagicDrive-V2, DriveDreamer, and Vista condition driving imagery on scene structure or control signals [11, 13, 14, 59]. GEN3C uses 3D information for camera-controlled video generation [45]. StreetCrafter and HorizonForge combine controllable diffusion with scene reconstruction or editing [60, 69]; WorldSplat generates Gaussians and refines their rendered videos [83], and DIFIX3D+ improves reconstructions with diffusion-based image refinement [62]. Perceptual similarity measures such as LPIPS assess rendered appearance [79]. Our formulation additionally assigns completed content a persistent identity, coordinate frame, and motion rule, so subsequent state changes reuse the same 3D completion.

3. Method

DriveCompose reconstructs a driving scene in the units that will later be controlled. We first recover visual evidence and aggregate it on background and canonical asset supports (Secs. 3.1–3.2). A typed codec defines the latent variables and their Gaussian decoding (Sec. 3.3). Conditional flow completes missing content in this representation (Sec. 3.4), after which the scene is composed and checked in the observed views (Sec. 3.5). Section 3.6 describes how these modules are learned. Figure 1 shows the complete construction and reuse path.

3.1. Problem formulation and compositional representation

The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}^r_{c_v}$. Rig poses $\mathbf{T}^w_r(t)$ are provided or estimated relative to the first frame, giving camera poses $\mathbf{T}^w_{c_v}(t) = \mathbf{T}^w_r(t)\mathbf{T}^r_{c_v}$. Transforms act on column vectors and distances are in meters. Images follow a rectified pinhole model and linear RGB; distortion, rolling shutter, and exposure differences are handled before evidence aggregation.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. A Gaussian stores its center $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$, opacity α, and directional appearance, and is rendered by Gaussian splatting [22]. Vehicles retain local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The decoder also carries forward the fixed support, identity, binding, and sky-validity metadata in Eq. 1. The layout is established from the input log; generation fills its unknown content. Simulation time changes the state of the decoded scene, while flow time indexes the construction of its latent codes.

3.2. Visual recovery and canonical aggregation

Shared visual encoding. The Driving Log Encoder uses patch tokenization [10] and transformer attention [54], with image-position, camera-calibration, and time embeddings. Within-image attention extracts spatial features. A second transformer processes pooled image summaries across views and timestamps, and returns the resulting context to the patch features. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. The confidence determines which predictions can contribute surface evidence.

Learned actor queries attend to all image tokens to form trajectory-level identity slots. Each slot then reads the features at an individual timestamp to estimate its state. This separates persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Root poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}_r^w(t)$. A human shape code selects an identity-specific template with fixed canonical support, skeleton topology, and binding. Slot capacity bounds the number of represented actors.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; vehicle anchors are motion-normalized and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \mathrm{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \mathrm{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The query confidence is the maximum over accepted observations. The test excludes observations whose depth or instance identity disagrees with the query. A reliable ray votes for free space only when the entire support cell is in front of the measured surface:

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{6}$$

Positive accumulated observation weight marks a query as observed. Otherwise, sufficient free-space votes mark it free, and the remainder is unknown. The masks m_i, f_i, and u_i are mutually exclusive and satisfy $m_i + f_i + u_i = 1$. Sky projection uses camera rotation alone, reliable sky semantics, and zero depth residual; sky queries receive no free-space votes.

3.3. Structured latent representation and decoding

Type-specific encoders map $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ to codes $\mathbf{z}_i \in \mathbb{R}^D$. Each query keeps its type, ownership, and coordinate frame even though all codes have the same dimension. Visibility masks and confidence are supplied separately to the completion model. Partial and complete observations therefore use the same intrinsic encoding map. Before conditioning, unobserved codes are set to zero; the observed subset is denoted $\mathbf{z}^{\text{obs}}$.

Background, vehicle, and human decoders independently map each query to K child Gaussians. For child k of query i, the center is confined to its support cell:

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{7}$$

Positive scales and normalized rotations determine positive-definite covariance. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. Human children inherit the skinning weights of their support query; vehicle children share their asset's rigid transform. A separate sky head predicts cubemap colors, with the observation-supported domain retained in V_{sky}.

Appearance uses a constant RGB term and three local directional coefficients per channel, evaluated as $c(d) = \max(0, c_0 + C_{\text{app}} \hat{d}_{\text{local}})$. Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can

exchange context across queries without changing this decoding property.

3.4. Conditional generative reconstruction

Scene-conditioned transformer. We parameterize the completion velocity with a DiT-style transformer [38] trained by flow matching [27]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. Shared memory then communicates their context:

$$\mathbf{M}' = \mathbf{M} + \mathrm{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{8}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \mathrm{Attn}_w(\mathrm{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{9}$$

Here $\widetilde{\mathbf{X}}$ concatenates locally updated tokens, and attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters the local normalization; learned tokens and log summaries initialize memory. This structure keeps each support's local organization while allowing a vehicle or human query to use information from the surrounding scene.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\mathrm{obs}} + u \odot [(1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{10}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\mathrm{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{11}$$

All unknown queries interpolated in Eq. 10 require valid target codes, even if s_i excludes some of them from the loss. This requirement determines how complete targets are constructed in Sec. 3.6.

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{12}$$

The projection $\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed fixes one sampled completion. Sampling precedes decoding, and the resulting scene is reused when the camera or actor states change.

3.5. Scene composition and observation consistency

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, vehicle transforms, and human root and joint poses. A vehicle's Gaussian center and covariance transform as

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{13}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. The transformed center is $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$. Its inverse $\mathbf{J}_i^{-1}$ maps viewing directions into the canonical frame; the direction is normalized before appearance evaluation. Nearly singular skinning transforms are rejected. The root pose controls overall placement, while the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. External policy and dynamics can advance the state:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{14}$$

$$s_{k+1} = F_{\mathrm{dyn}}(s_k, \pi(o_k), \Delta t). \tag{15}$$

Resetting or branching a trajectory changes state while preserving $\mathcal{W}$ and its seed. Finite support and supported sky directions determine coverage. Road meshes and collision proxies share the visual assets' coordinates and are maintained by the physical backend.

Checking a constructed scene. Fixed observed codes preserve decoded parameters, but a newly generated surface can still occlude them. Before accepting a scene, we render the completed candidate and an observed-only reference at the original cameras and actor states. The reference disables all non-observed queries. Let $e_{\mathrm{cand}}(p)$ and $e_{\mathrm{obs}}(p)$ be their mean absolute RGB errors at pixel p. We measure local degradation with

$$\Delta e(p) = \max\{e_{\mathrm{cand}}(p) - e_{\mathrm{obs}}(p), 0\}. \tag{16}$$

Averaging this quantity over reliable non-ego pixels prevents improvements in one region from cancelling degradation elsewhere. Acceptance also bounds absolute RGB and valid-depth errors, minimum coverage, and foreground intrusion, using aggregate and worst-view checks. An intrusion is a newly rendered surface in front of the predicted reference depth that is absent from the observed-only render; its rate is normalized over reliable non-sky geometry pixels. Predicted depth provides an observation-consistency reference. Independent geometry evaluation requires a separate reference. A rejected candidate is not exported for reuse.

3.6. Training procedure

Learning is divided into four stages so that the completion target has a stable meaning. Recovery determines the evidence and support, the codec defines its latent coordinates, and the completion network learns a distribution over the missing codes. The last stage adapts this distribution to real observations without changing the representation it predicts.

Stage 1: visual recovery. We supervise the image encoder, geometry heads, identity slots, and state heads using geometry, semantics, instance assignments, and actor annotations. One trajectory-level matching assigns slots to target identities and is reused at every timestamp. Root translation, rotation, and fixed dimensions constrain vehicle states; humans additionally require identity-consistent shape, template binding, and joint supervision. Confidence is trained or calibrated against geometric reliability, using depth, occlusion, and reprojection information. These targets establish the coordinate and identity consistency required by aggregation.

Stage 2: latent representation fitting. We freeze recovery and train the type-specific encoders and decoders. For each sample, partial input establishes the query layout once. More complete observations are then aggregated on exactly the same positions, types, order, and ownership. This correspondence is required for latent completion: separately rebuilding a complete-scene layout would change the variables being predicted. RGB, depth, and silhouette supervision train the decoded scene, with empty-space opacity and Gaussian-scale regularization. Gaussian-set targets can also be matched as sets; direct coordinate supervision is used only when child correspondences are available.

For rendered image $\widehat{\mathbf{I}}$ and depth $\widehat{d}$, the observation terms used in representation fitting and later adaptation are masked L1 errors:

$$\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \tag{17}$$

$$\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \tag{18}$$

The notation $\langle \ell \rangle_\Omega$ denotes a mean over valid elements of mask Ω. The depth target d and its validity mask are supplied by the available supervision; they do not make the observation check an independent accuracy measurement.

Stage 3: completion-prior learning. We freeze recovery and the entire codec, encode complete targets as $\mathbf{z}_1$, and optimize the completion transformer with Eq. 11. Complete scene or asset examples must provide a valid target for every unknown query in Eq. 10; partial real logs alone do not meet this condition. Complete scenes provide cross-component context, while complete assets supply within-object content. A validity test rejects targets with unsupported unknown queries.

Conditioning masks simulate contiguous unseen surfaces, background holes, long occlusions, short tracks, and uneven camera coverage. Perturbations to recovered poses, instance assignments, and confidence expose the generator to imperfect evidence. Target codes and input codes retain the same codec and query definition throughout this stage.

Stage 4: adaptation through rendering. Keeping recovery and the codec frozen, we differentiate through flow integration, decoding, and rendering to update only the completion network. Partial real logs supervise the rendered observations. Known-free rays and geometric backend samples constrain added content:

$$\mathcal{L}_{\text{scene}} = \lambda_e \langle a_r^2 \rangle + \lambda_g \langle [-d_f]_+^2 \rangle + \lambda_c \langle |d_f| \rangle_{\Omega_c} + \lambda_p \langle [d_p]_+^2 \rangle, \tag{19}$$

$$\mathcal{L}_{\text{adapt}} = \lambda_r \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scene}}. \tag{20}$$

Here a_r is accumulated opacity along a known-free ray, d_f is foot signed distance positive above ground, Ω_c selects contacting feet, and d_p is penetration depth positive inside another actor. We use $[x]_+ = \max(x, 0)$; unmasked means run over supplied samples. The coefficients λ weight the active constraints. Free-ray opacity constrains the rendered effects of added content beyond the query-level free-space mask.

Complete-target samples can be interleaved with real-log adaptation to retain the learned prior, yielding $\mathcal{L}_4 = \lambda_f \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{adapt}}$, with each term applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration.

Inference. A log is encoded once, its support and evidence are fixed, and the completion model integrates the unknown codes. We then decode and check the scene before exporting its fixed content. Runtime rendering uses only this content and the requested states. Recovery, codec, completion, and query-definition versions are recorded together, since changing any of the first two changes the latent distribution on which completion was learned.

4. Experiments

We evaluate our method along four axes: *(i)* dynamic scene reconstruction quality on Waymo Open Dataset, covering temporal interpolation and extrapolation (§ 4.2); *(ii)* off-trajectory view synthesis under lateral camera shifts (§ 4.3); *(iii)* cross-dataset transfer from Waymo to nuScenes (§ 4.4); and *(iv)* ablations of the main design choices (§ 4.5).

4.1. Experimental Setup

Datasets and protocol. We use the Waymo Open Dataset with 798 training and 202 validation sequences. Following STORM, each validation sequence is divided into non-overlapping 2-second clips of 20 frames at 10 Hz [71]. The FRONT, FRONT_LEFT, and FRONT_RIGHT cameras provide observations at zero-based frame indices $\{0, 5, 10, 15\}$. The remaining 16 frames are used for evaluation: 12 frames for temporal interpolation and four frames for prediction beyond the last observation. Predictions and reference images are resized to 240×160 pixels for evaluation [21].

Baselines. We compare with STORM, DGGT, StreetForward [5, 71, 77]. Methods use the same context timestamps and target views, with model-specific preprocessing. Camera parameters, additional geometric inputs, and rendering adapters are documented with the implementation settings. For spatial extrapolation, we compare Gaussian renderings before diffusion refinement, following PointForward [8].

Metrics. We report PSNR and SSIM over the full image and shared annotated regions. Depth quality is measured using AbsRel and δ_1 at valid projected LiDAR samples after per-frame scale alignment [21]. Scores are averaged over frames within each clip and then over clips within each sequence; sequences receive equal weight. Novel-trajectory renderings are evaluated using FID and FVD against a fixed real-image and video reference population.

Implementation details. DriveCompose reconstructs category-specific actor assets from motion-aligned observations and composes them with a static background. Reconstruction time includes preprocessing, asset generation, and scene composition. Rendering time is measured separately, and asset reuse is evaluated across repeated scene queries.

4.2. Dynamic Reconstruction and Temporal Prediction

We evaluate reconstruction at unobserved times along the recorded camera trajectory. Table 1 reports image and depth quality. The regional comparison focuses on actor appearance, and synchronized visualizations examine object position, shape, and occlusion changes.

Temporal interpolation and prediction. Table 2 separates temporal interpolation from prediction at 0.1–0.4 seconds beyond the last context frame.

Table 1. Dynamic scene reconstruction on Waymo. Image quality is reported over the full image and annotated regions; depth is evaluated at valid LiDAR samples.

	Full image		Dynamic Region		Depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.191	0.833	22.083	0.654	0.129	0.867
DGGT	26.056	0.809	20.809	0.603	0.168	0.779
StreetForward	25.256	0.814	21.844	0.636	0.122	0.860
DriveCompose	–	–	–	–	–	–

Table 2. Temporal interpolation and extrapolation on Waymo. PSNR and SSIM are reported separately for interpolation and extrapolation; context frames are excluded.

	Interp.		Extrap.		All targets	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
STORM	26.427	0.842	25.481	0.807	26.191	0.833
DGGT	26.265	0.817	25.429	0.788	26.056	0.809
StreetForward	25.653	0.827	24.065	0.776	25.256	0.814
DriveCompose	–	–	–	–	–	–

Table 3. Novel-view synthesis under lateral camera shifts. FID and FVD are averaged over left and right shifts at each displacement magnitude. Lower is better.

Method	Shift ±1 m		Shift ±2 m		Shift ±4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
DriveCompose	–	–	–	–	–	–

4.3. Off-Trajectory View Synthesis

We shift the camera trajectory laterally by 1, 2, and 4 meters to the left and right, keeping camera orientation, intrinsics, and timestamps fixed. This evaluation tests the stability of scene structure and the rendering of newly visible surfaces as the viewpoint moves away from the observations [8, 83].

Table 3 reports FID and FVD at each displacement magnitude, averaged over separately evaluated left and right shifts. Both metrics use the same real reference population across methods. Qualitative comparisons use matched scenes, timestamps, and camera offsets to examine vehicle shape, road layout, and occlusion boundaries.

4.4. Cross-Dataset Generalization

We evaluate transfer from Waymo to nuScenes under zero-shot and fine-tuning settings. Zero-shot evaluation di-

Table 4. Cross-dataset generalization and target-domain adaptation. Zero-shot models are evaluated directly on the target datasets, while trained models are fine-tuned on the target domain. Higher PSNR and SSIM, and lower LPIPS are better.

	Full image			Dynamic regions		
Method	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Zero-shot						
STORM	–	–	–	–	–	–
DGGT	–	–	–	–	–	–
StreetForward	–	–	–	–	–	–
DriveCompose	**–**	**–**	**–**	**–**	**–**	**–**
Trained						
STORM	–	–	–	–	–	–
DriveCompose	**–**	**–**	**–**	**–**	**–**	**–**

Table 5. **Component ablations.** All variants use $T = 8$, a shared recovery module and codec, and the same evaluation masks. Quality is measured before acceptance filtering; Pass includes all requested worlds. A1 requires no completion training. Numerical cells are left blank.

Variant	New-MAE ↓	Obs-Reg ↓	Cov. (%) ↑	Pass (%) ↑
Full (A0)				
No completion (A1)				
Independent (A2)				
No world coord. (A3)				
Soft obs. (A7)				

rectly applies the Waymo-trained checkpoint without target-domain updates. Fine-tuning adapts the same initialization using the nuScenes training split and evaluates it on the held-out validation split. Both settings use the same target evaluation scenes, context sampling, and metrics.

Table 4 reports the two settings separately. The zero-shot comparison assesses transfer to new environments, while the fine-tuning comparison measures adaptation to the target domain. Full-image and annotated-region scores distinguish overall scene quality from actor reconstruction.

4.5. Ablation protocol

Inputs and controls. Paired evaluation of hidden surfaces is defined on independent complete-reference scenes; Waymo logs provide recorded-view evaluation. Each 29-timestamp window uses three synchronized cameras: front, front-left, and front-right. The nested input sets are $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$, with shared target timestamps $\{2, 6, \ldots, 26\}$. Target images are excluded from evidence and query construction; target camera and actor states are supplied equally to every variant. The recovery module and codec are frozen, with matched query layouts, data exposure, and optimization budgets at each input setting. Training samples $T \in \{2, 4, 8\}$ uniformly; completion uses 24 Euler steps. The four learned variants use training seeds $\{0, 1, 2\}$ and world seeds $\{7, 17, 27\}$; A1 is deterministic. Hyperparameters and acceptance thresholds are selected on an internal development split before evaluation on scene- and identity-disjoint data.

Metrics. New-MAE measures linear-RGB error on independently identified surfaces unseen in every input; NVS-MAE covers all held-out target pixels. Across input settings, New-MAE uses the common region unseen even under $\mathcal{T}_8$. Obs-Reg averages the positive input-view regression in Eq. 16. Cov. is the non-sky target fraction with support coverage > 0.5, opacity > 0.1, and finite positive depth. Uncovered pixels receive the same zero-RGB fill for error measurement. Pass is the fraction of all requested worlds accepted by the fixed checker, including construction failures in its denominator. Image metrics include every renderable candidate, including rejected worlds; non-renderable counts are recorded separately. Metrics are averaged within each scene and then across scenes and training seeds.

Figure 2 reserves matched crops for completion and observation preservation, followed by three state interventions from the same world.

4.6. Ablation studies

Completion and scene context. Table 5 compares Full (A0) with three controls. No completion (A1) disables the opacity of every non-observed, non-sky query. Independent (A2) isolates component memories at every layer while sharing parameters and retaining the same global log context. No world coord. (A3) zeros only the explicit reference-world coordinate channels in the completion network; canonical geometry and rendering transforms remain unchanged. The learned controls are retrained through Stages 3–4; memory-related compute differences are recorded.

Observation preservation. Soft observation (A7) allows observed codes to evolve instead of hard resetting them. Its observed flow path is $(1 - \tau)\mathbf{z}^{\text{obs}} + \tau\mathbf{z}_1$, with target velocity $\mathbf{z}_1 - \mathbf{z}^{\text{obs}}$. Observed and unknown flow losses are separately normalized and equally weighted. Rendering adaptation adds a mean-squared penalty on observed-code changes, with its weight selected from $\{0.1, 1, 10\}$ on development data. Observed conditioning, free-space constraints, the codec, and the checker remain fixed. This comparison examines the tradeoff between completion and input-view preservation.

Input sparsity. Table 6 reuses the A0/A2 checkpoints at $T = 2, 4, 8$, with 6/12/24 input images. The time span, tar-

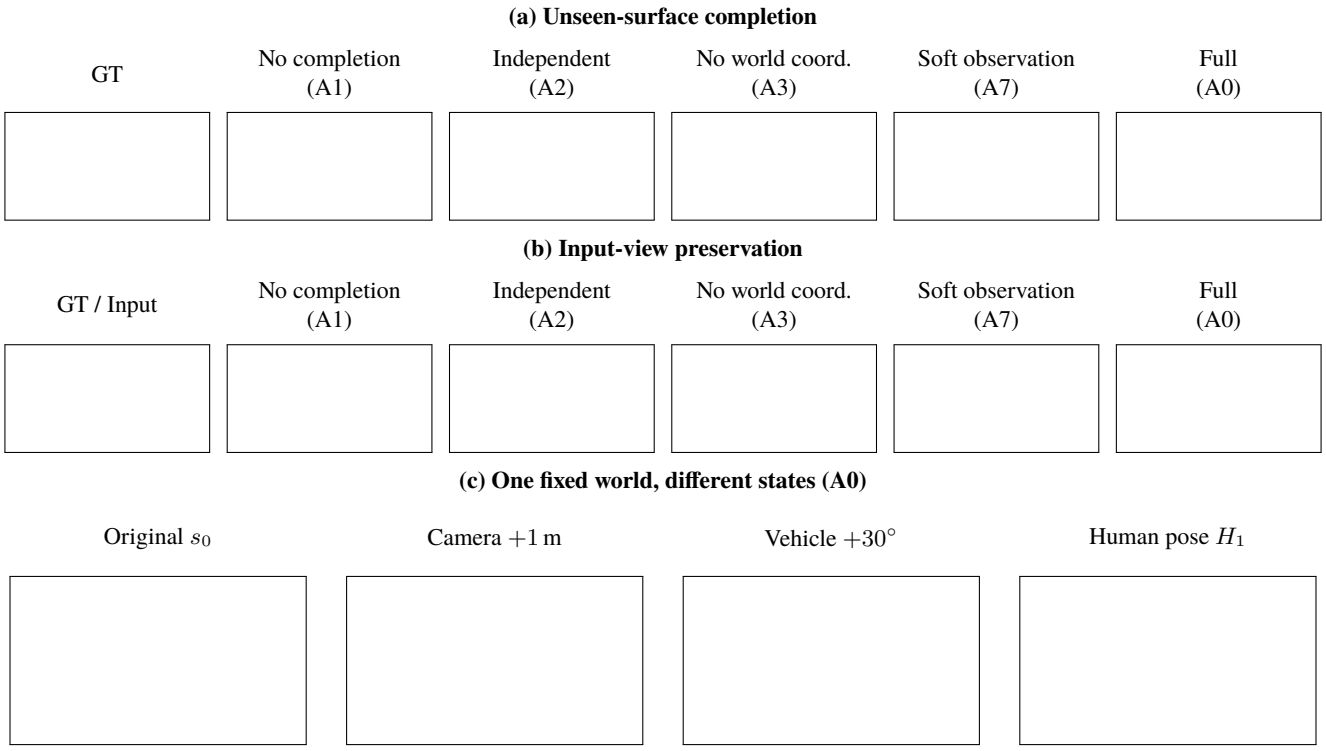


Figure 2. **Ablation and reuse layout.** (a) The same newly exposed region across five variants and an independent reference. (b) Matched crops in an original input view. (c) Independent camera, vehicle, and human-state changes from one A0 world with seed 7; the other states and lighting are fixed. All image slots are intentionally empty.

Table 6. **Input-timestamp sensitivity.** Three cameras and a fixed time span are shared across the nested input sets. The same checkpoints and target images are used at every T; New-MAE uses the common unseen region. Numerical cells are left blank.

Variant	T	NVS-MAE ↓	New-MAE ↓	Cov. (%) ↑
Full (A0)	2			
Independent (A2)	2			
Full (A0)	4			
Independent (A2)	4			
Full (A0)	8			
Independent (A2)	8			

get views, and common unseen-region mask remain fixed; evidence and queries are rebuilt from each input subset. No model is retrained for this comparison.

4.7. State-driven reuse

At timestamp 14 and world seed 7, the interventions independently translate the camera by 1 m along its local $+x$ axis, rotate a vehicle by $30°$ about its root and the world vertical axis, or apply a held-out valid human pose H_1 with fixed root placement. Lighting and canonical assets remain fixed, without resampling completion.

5. Limitations

DriveCompose represents a finite region under fixed illumination. Errors in recovered depth, identity, or canonical pose can enter the observed anchors and constrain later completion incorrectly. Vehicles are rigid, and humans retain fixed clothing volume and skeleton binding; poses outside the template's support can expose missing geometry. Reference-world coordinates provide context at construction, while contact and collision validity after an edit depend on the physical backend. The observation check assesses agreement with recorded evidence. Geometry outside that evidence and appearance under changed illumination require separate supervision or modeling.

6. Conclusion

We presented DriveCompose, a generative reconstruction framework for compositional driving scenes. Its structured latent representation gathers image evidence into background and canonical assets, giving missing content a defined location, identity, and motion model before generation. Conditional completion uses scene context while retaining observed evidence, and typed decoding produces the components of a reusable Gaussian scene. Separating content construction from state-driven composition allows sub-

sequent camera and actor changes to reuse the same completion.

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